Flame and its structure



Which is the festival of light? What is the specialty of that festival? Yes. We will light more lamps to decorate the houses. Won't we? Now how do the lights glow? Yes, with flame.



Here is an experiment with colorful flame

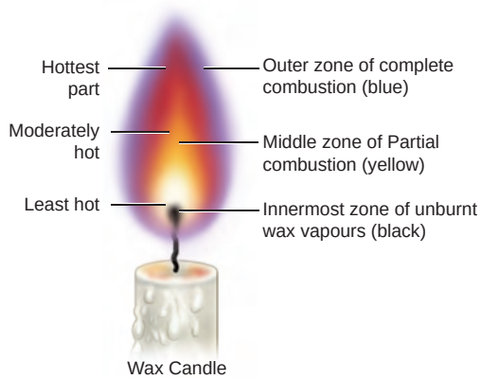
- White flame – Epsom salt ($MgSO_4$)
- Violet flame – Lithium Chloride
- Indigo flame – Potassium Chloride
- Blue flame – Bleaching powder
- Green flame – Borax powder
- Yellow flame – Calcium chloride
- Orange flame – Table salt
- Red – Strontium chloride

Teacher shows the experiment with these salts soaked in alcohol and makes fire.

Flame:

Flame is a zone of combustions of a combustible substance. Substances which vaporize during burning produce flames. E.g. Wax, Kerosene etc.

Substances which do not vaporize during burning do not produce flames e.g. coal.



Structure of a Candle flame

A candle flame has three main zones, they are

i. The outer zone – complete combustion of the fuel takes place and the colour of the flame is blue and is the hottest part of the flame. It is the non-luminous part of the flame.

ii. The middle zone -partial combustions of the fuel takes place and the colour of the flame is yellow and is moderately hot part of the flame. It is the luminous part of the flame.

iii. The inner zone: There are unburnt vapours of the fuel and the colour is black and is least hot part.

Finger inside the flame



Why is the candle flame straight?



DO YOU KNOW?

A candle flame is caused by vapour burning above the candle. This burning vapour is hotter than the surrounding air and is therefore less dense. So, by the principle of convection, it "rises" so the flame is always upwards

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